CCNA – Module 4: Routing and Switching | Assessment M4-A1

Name: ABHIJITH NAIR

Recommended Packet Tracer Addressing Plan

Device / Interface	VLAN	IP Address	Subnet Mask	Default Gateway
PC1 – Sales	10	192.168.100.10	255.255.255.0	192.168.100.1
PC2 – Support	20	192.168.200.10	255.255.255.0	192.168.200.1
Router G0/0.10	10	192.168.100.1	255.255.255.0	—
Router G0/0.20	20	192.168.200.1	255.255.255.0	—

Note: If the router uses GigabitEthernet0/0/0 instead of GigabitEthernet0/0, substitute G0/0/0 wherever G0/0 appears below.

SECTION A – CONCEPT APPLICATION

S1. Static Routing vs Dynamic Routing

In static routing, every route is entered by the administrator by hand; the router never discovers paths from its neighbours. In dynamic routing, protocols such as RIP and OSPF let routers advertise networks to each other and build their tables automatically.

Static Routing	Dynamic Routing
Routes are entered manually by the administrator.	Routes are discovered automatically via routing protocols.
Consumes almost no CPU, memory or link bandwidth.	Consumes router resources and bandwidth to send updates.
Any topology change requires manual reconfiguration.	Automatically recalculates paths when the topology changes.
Ideal for small, stable networks.	Suited to medium and large networks.
Generates no protocol traffic on the wire.	Constantly exchanges routing updates between routers.

Why static routing fits this scenario:

- The company operates only two routers, so the number of routes to maintain is tiny.
- The network layout almost never changes, so automatic recalculation adds no value.
- Static routes are quick to configure and easy to troubleshoot.
- There is no routing-update overhead, keeping resource usage minimal.
- In a small, stable environment, static routing is more efficient than running a dynamic protocol.

Example static route: ip route 192.168.200.0 255.255.255.0 <next-hop>

S2. RIP vs OSPF

Feature	RIP	OSPF
Routing type	Distance Vector	Link State
Metric	Hop count	Cost (bandwidth based)
Maximum hop count	15 usable hops	No hop-count limit
Convergence speed	Slow	Fast
Scalability	Very small networks	Medium to large networks
Algorithm	Bellman-Ford	Dijkstra SPF
Administrative Distance	120	110

Recommended protocol: OSPF

- OSPF reconverges quickly after a link or route failure.
- It scales well as new branch routers are added over time.
- RIP is restricted to 15 hops and suits only very small networks.
- OSPF builds a complete topology map and computes the best path with the SPF algorithm.
- For a five-branch network that keeps growing, OSPF is the better long-term option.

S3. MAC Address Table and Frame Forwarding

A switch builds its MAC address table dynamically by inspecting the source MAC address of each incoming frame and recording that address against the port it arrived on.

- PC1 transmits a frame into the switch on port Fa0/1.
- The switch reads the source MAC, and learns that PC1 is reachable via Fa0/1.
- If the destination MAC is still unknown, the switch floods the frame (unknown unicast) out every other port in the same VLAN except the receiving port.
- PC2 gets the frame and responds through Fa0/2.
- The switch now learns PC2's MAC address on Fa0/2.
- From this point onward, frames between the two PCs are forwarded only out the correct port, not flooded.

Learned Device	Switch Port
PC1 MAC Address	Fa0/1
PC2 MAC Address	Fa0/2

Simple rule: learn the source, look up the destination, then forward to one port or flood if the destination is unknown.

S4. VLAN Segmentation and 802.1Q Trunking

A VLAN divides one physical switch into multiple logical broadcast domains. For instance, Sales devices can sit in VLAN 10 while Support devices sit in VLAN 20, even though they share the same hardware.

Key benefits of VLAN segmentation:

- Better security through department separation.
- Less unnecessary broadcast traffic on each segment.
- Cleaner network organisation and easier management.
- Users can be grouped logically regardless of physical location.
- Improved performance because broadcasts stay within their VLAN.

An 802.1Q trunk transports frames from several VLANs across a single physical link. The transmitting switch inserts a VLAN tag into each frame on the trunk so the device at the far end knows which VLAN the frame belongs to. This lets VLAN 10 and VLAN 20 traffic share one switch-to-router link while remaining logically separate.

S5. Inter-VLAN Routing and Router-on-a-Stick

Hosts in different VLANs live in different Layer 3 subnets, so a Layer 2 switch alone cannot pass traffic between them. Inter-VLAN routing is needed to move packets between VLAN 10 and VLAN 20.

Router-on-a-Stick solves this with a single physical interface divided into logical sub-interfaces. Each sub-interface is tagged with an 802.1Q VLAN ID and given an IP address that serves as the default gateway for that VLAN:

G0/0 → G0/0.10 (VLAN 10, 192.168.100.1/24) and G0/0.20 (VLAN 20, 192.168.200.1/24)

Why Router-on-a-Stick beats one physical interface per VLAN:

- Only one physical router port is consumed.
- Less cabling and fewer router interfaces are required.
- More cost-effective for a small startup.
- New VLANs can be supported by simply adding sub-interfaces.
- Easier to expand than dedicating hardware ports to every VLAN.

S6. Wireless Security and MAC Filtering

Wireless networks should rely on strong authentication and encryption. Open networks and WEP are considered insecure and must be avoided. WPA2 or WPA3 is the minimum standard, with the Enterprise mode preferred where centralised authentication is available.

Option	**Use / Security Level**
Open Wi-Fi	No authentication; never recommended.
WEP	Outdated and easily cracked; not recommended.
WPA/WPA2 Personal	Shared passphrase; acceptable for small offices.
WPA3	Stronger, modern protection where devices support it.
WPA2/WPA3 Enterprise	Per-user credentials; best for controlled business access.

MAC Filtering	Permits or denies devices by MAC address; only a supplementary control.

Recommended approach: deploy WPA2/WPA3 Enterprise so every employee gets unique credentials. This gives strong access control while keeping onboarding manageable. MAC filtering may be enabled as a secondary layer, but it must never be the primary defence because MAC addresses can be spoofed. In a very small office without an authentication server, use WPA2/WPA3 Personal with a strong passphrase, optionally combined with MAC filtering.

SECTION B – PRACTICAL CONFIGURATION TASKS

Recommended topology:

PC1 (Fa0) —— Switch Fa0/1 PC2 (Fa0) —— Switch Fa0/6 Switch Fa0/23 —— Router G0/0

Use one Cisco 2960 switch, one Cisco router (1941 or 2911) and two PCs, connected with copper straight-through cables.

Task 1 – Configure Sales and Support VLANs

Requirement: create VLAN 10 named Sales and VLAN 20 named Support, assign access ports, and verify.

Switch configuration:

```
enable
configure terminal
vlan 10
 name Sales
 exit
vlan 20
 name Support
 exit
interface range fa0/1-5
 switchport mode access
 switchport access vlan 10
 exit
interface range fa0/6-10
 switchport mode access
 switchport access vlan 20
 exit
end
copy running-config startup-config
```

Verification command: show vlan brief

Expected result: VLAN 10 lists the Sales access ports and VLAN 20 lists the Support access ports. Capture a screenshot of the output.

Task 2 – Configure an 802.1Q Trunk Link

Assume switch port Fa0/23 connects to router interface G0/0.

```
enable
configure terminal
interface fa0/23
 switchport mode trunk
 switchport trunk allowed vlan 10,20
 end
copy running-config startup-config
```

Verification commands: show interfaces trunk | show interface fa0/23 switchport

- Fa0/23 should report a trunking status.
- The allowed VLAN list should contain only VLANs 10 and 20.
- Keep the native VLAN identical at both ends; leave it at the default (VLAN 1) in this basic topology.

Task 3 – Router-on-a-Stick for Inter-VLAN Routing

Router configuration:

```
enable
configure terminal
interface g0/0
 no shutdown
 exit
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.100.1 255.255.255.0
 exit
interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.200.1 255.255.255.0
 exit
end
copy running-config startup-config
```

PC1 IP settings: IP 192.168.100.10 Mask 255.255.255.0 Gateway 192.168.100.1

PC2 IP settings: IP 192.168.200.10 Mask 255.255.255.0 Gateway 192.168.200.1

Verification: show ip interface brief — the physical interface and both sub-interfaces must be up/up. Then test:

PC1> ping 192.168.100.1
PC2> ping 192.168.200.1
PC1> ping 192.168.200.10

A successful final ping from PC1 to PC2 proves inter-VLAN routing works. Capture a screenshot.

Task 4 – Capture and Analyse Inter-VLAN Traffic

Generate ICMP traffic from the VLAN 10 host to the VLAN 20 host: ping 192.168.200.10

Useful Wireshark display filters: icmp | vlan && icmp

Traffic path: PC1 (192.168.100.10 / VLAN 10) → Switch → 802.1Q Trunk → Router → 802.1Q Trunk → Switch → PC2 (192.168.200.10 / VLAN 20)

Frame analysis:

- PC1 builds an IP packet with source 192.168.100.10 and destination 192.168.200.10.
- Because the destination sits on another subnet, PC1 addresses the Ethernet frame to its default gateway — the destination MAC is the router's MAC, not PC2's.
- As the frame crosses the switch-to-router trunk, it carries an 802.1Q tag for VLAN 10.
- The router strips the incoming Layer 2 header, reads the destination IP, selects VLAN 20, and builds a brand-new Ethernet frame.
- The outgoing frame uses the router interface MAC as source and PC2's MAC as destination, and is tagged for VLAN 20 on the trunk.
- The switch removes the tag before delivering the frame out PC2's access port.

Location	Source IP	Destination IP	Source MAC	Destination MAC	VLAN
PC1 → Switch	192.168.100.10	192.168.200.10	PC1 MAC	Router MAC	10
Switch → Router	192.168.100.10	192.168.200.10	PC1 MAC	Router MAC	10 (tagged)
Router → Switch	192.168.100.10	192.168.200.10	Router MAC	PC2 MAC	20 (tagged)
Switch → PC2	192.168.100.10	192.168.200.10	Router MAC	PC2 MAC	20

Report explanation: the source and destination IP addresses stay identical end-to-end, but the Layer 2 MAC addresses are rewritten at the router because the router removes the old frame and builds a new one for the destination subnet. The 802.1Q tag on the trunk simply identifies whether the frame belongs to VLAN 10 or VLAN 20.

SECTION C – MINI CAPSTONE PROJECT

TechLaunch Startup Network Build and Fault Diagnosis

Objective: build a two-VLAN network for Sales and Support, configure an 802.1Q trunk and Router-on-a-Stick, then resolve a fault where Sales hosts can talk inside their own VLAN but cannot reach the Support file server.

Network Plan

Department	VLAN	Network	Gateway	Example Host
Sales	10	192.168.100.0/24	192.168.100.1	192.168.100.10
Support	20	192.168.200.0/24	192.168.200.1	192.168.200.10

Switch Configuration

```
enable
configure terminal
vlan 10
 name Sales
 exit
vlan 20
 name Support
 exit
interface range fa0/1-5
 switchport mode access
 switchport access vlan 10
 exit
interface range fa0/6-10
 switchport mode access
 switchport access vlan 20
 exit
interface fa0/23
 switchport mode trunk
 switchport trunk allowed vlan 10,20
 no shutdown
 exit
end
copy running-config startup-config
```

Router Configuration

```
enable
configure terminal
interface g0/0
 no shutdown
 exit
```

```
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.100.1 255.255.255.0
 exit
interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.200.1 255.255.255.0
 exit
end
copy running-config startup-config
```

Ping-Based Fault Isolation Process

- From the Sales host, ping another device in VLAN 10. Success proves local VLAN switching works.
- Ping the Sales gateway 192.168.100.1. If this fails, inspect the access VLAN assignment, the trunk, or the router's physical interface.
- Ping the Support gateway 192.168.200.1. If the Sales gateway replies but this does not, check the Router-on-a-Stick sub-interfaces and VLAN tagging.
- Ping the Support file server 192.168.200.10. If both gateways reply but the server does not, verify the server's IP configuration, gateway, access-port VLAN, and cabling.
- Use the commands below to establish whether the fault is Layer 2 or Layer 3.

Layer 2 verification (switch): show vlan brief | show interfaces trunk | show interface fa0/23 switchport | show mac address-table

Layer 3 verification (router): show ip interface brief | show ip route | show running-config interface g0/0 | show running-config interface g0/0.10 | show running-config interface g0/0.20

Example Root Cause and Fix

Root cause: the physical router interface G0/0 was left administratively down because the no shutdown command was omitted. Since every Router-on-a-Stick sub-interface depends on its parent interface, inter-VLAN routing fails even though same-VLAN traffic still flows.

Verify before fixing: show ip interface brief — the parent interface shows as administratively down.

Fix:

```
enable
configure terminal
interface g0/0
 no shutdown
end
```

Verify the repair: show ip interface brief (all interfaces up/up), then repeat the pings.

Short Written Capstone Report

The network was built with VLAN 10 for Sales and VLAN 20 for Support. Access ports were placed in the correct VLANs and Fa0/23 was configured as an 802.1Q trunk carrying VLANs 10 and 20. Router-on-a-Stick was implemented with sub-interfaces G0/0.10 and G0/0.20. During testing, same-VLAN communication succeeded but inter-VLAN traffic failed. The command show ip interface brief revealed that the router's physical G0/0 interface was administratively down. The missing no shutdown command was the root cause; after adding it, pings between VLANs succeeded.

SECTION D – AI-AUGMENTED LEARNING

Step 1 – Prompt Given to the AI Tool

"Create a Cisco Packet Tracer configuration for a startup with these requirements:

- Create VLAN 10 named Sales.
- Create VLAN 20 named Support.
- Assign switch ports Fa0/1–Fa0/5 to VLAN 10.
- Assign switch ports Fa0/6–Fa0/10 to VLAN 20.
- Configure Fa0/23 as an 802.1Q trunk toward a router.
- Allow only VLANs 10 and 20 on the trunk.
- Configure Router-on-a-Stick on router interface G0/0.
- Configure G0/0.10 for VLAN 10 with IP 192.168.100.1/24.
- Configure G0/0.20 for VLAN 20 with IP 192.168.200.1/24.
- Ensure hosts in VLAN 10 and VLAN 20 can communicate.
- Provide the Cisco IOS commands and verification steps."

Step 2 – AI Original Configuration Script

Original switch configuration:

```
enable
configure terminal
vlan 10
 name Sales
 exit
vlan 20
 name Support
 exit
interface range fa0/1-5
 switchport mode access
 switchport access vlan 10
```

```
 exit
interface range fa0/6-10
 switchport mode access
 switchport access vlan 20
 exit
interface fa0/23
 switchport mode trunk
 switchport trunk allowed vlan 10,20
 no shutdown
 exit
end
```

Original router configuration produced by the AI:

```
enable
configure terminal
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.100.1 255.255.255.0
 exit
interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.200.1 255.255.255.0
 exit
end
```

Bug Found Without AI

The AI-generated router configuration never enabled the parent physical interface G0/0. The missing commands are:

```
interface g0/0
 no shutdown
```

If the parent interface stays administratively down, none of its sub-interfaces can carry traffic, which breaks inter-VLAN communication entirely.

Corrected Configuration

```
enable
configure terminal
interface g0/0
 no shutdown
 exit
interface g0/0.10
```

```
 encapsulation dot1Q 10
 ip address 192.168.100.1 255.255.255.0
 exit
interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.200.1 255.255.255.0
 exit
end
copy running-config startup-config
```

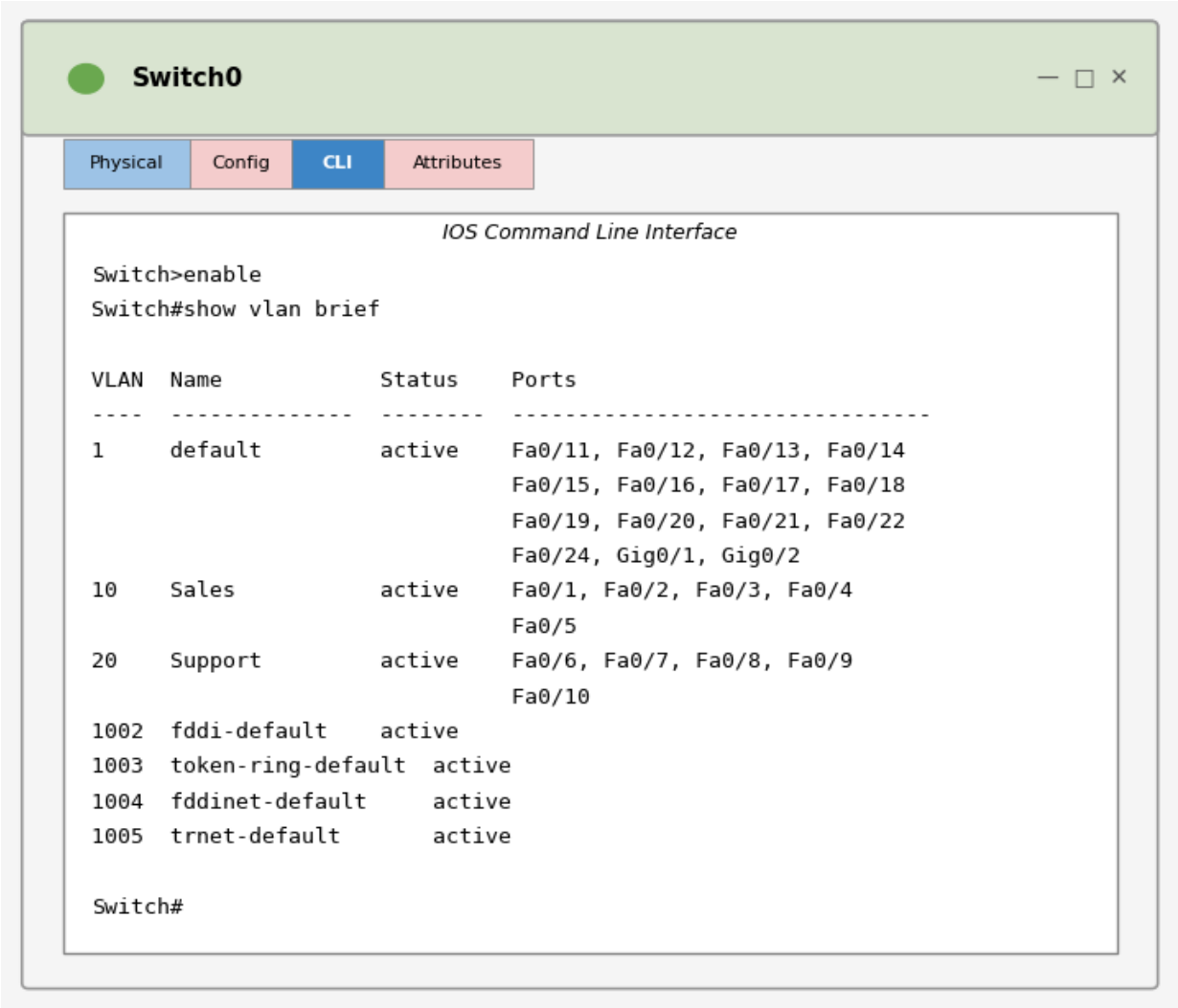
Verification: show ip interface brief | show vlan brief | show interfaces trunk | ping 192.168.200.10

Debugging Note

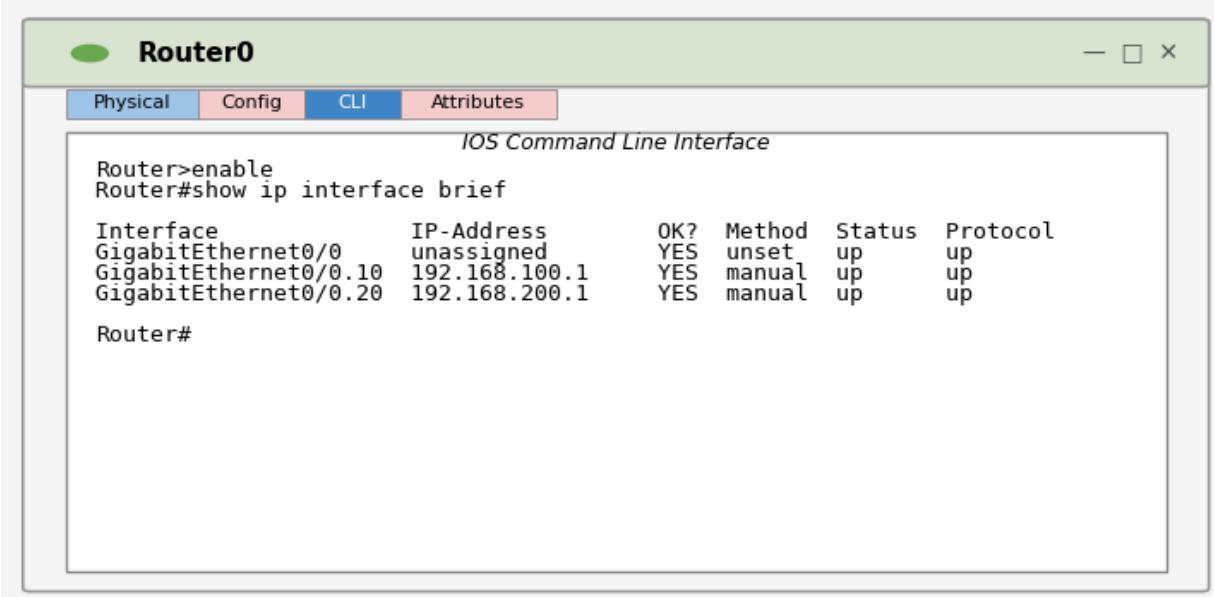
The AI-generated configuration correctly built the VLANs, the trunk, and the router sub-interfaces, but it never brought up the physical G0/0 interface. With the parent interface administratively down, sub-interfaces G0/0.10 and G0/0.20 could not forward any traffic. Adding no shutdown under G0/0 resolved the issue, and inter-VLAN pings then succeeded.

FINAL SCREENSHOTS CHECKLIST

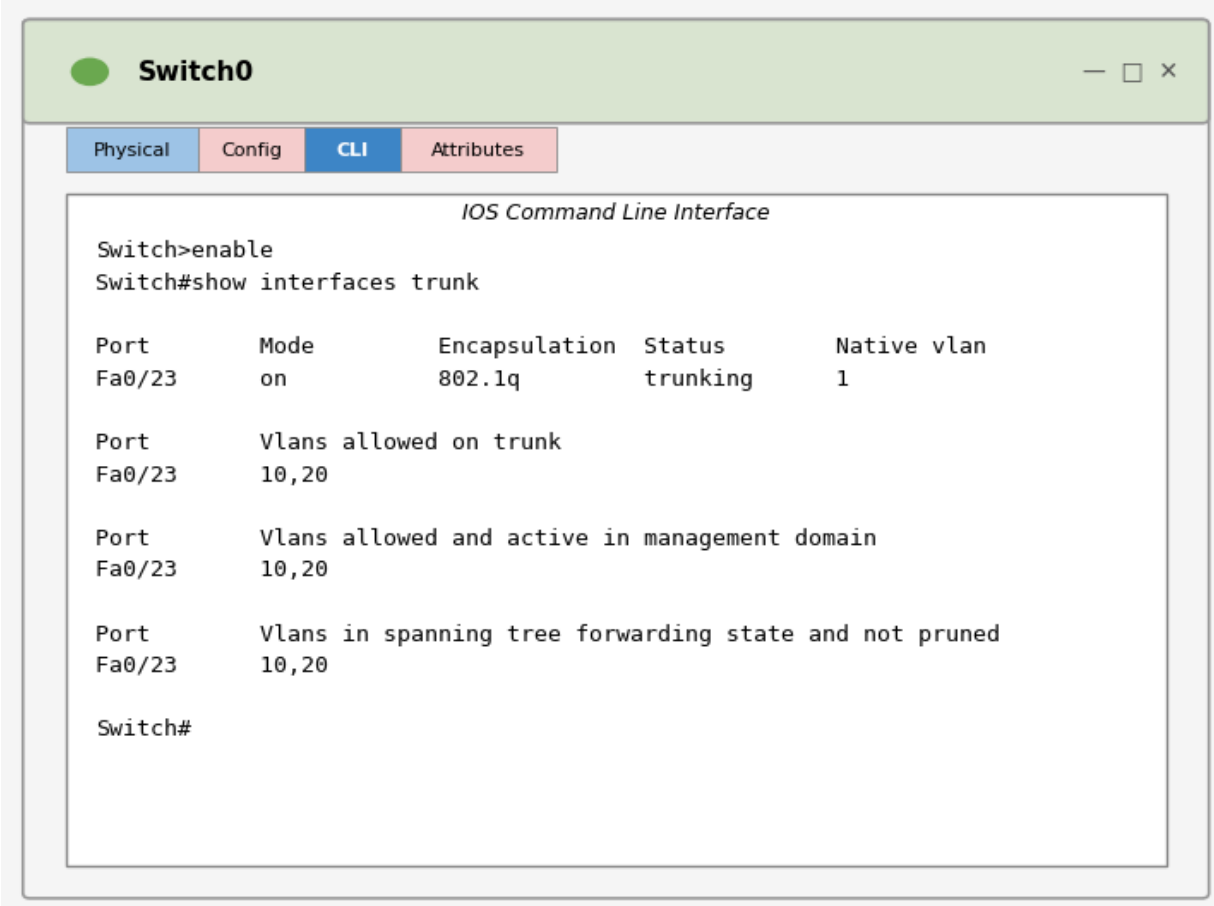
1. VLAN Configuration (show vlan brief)



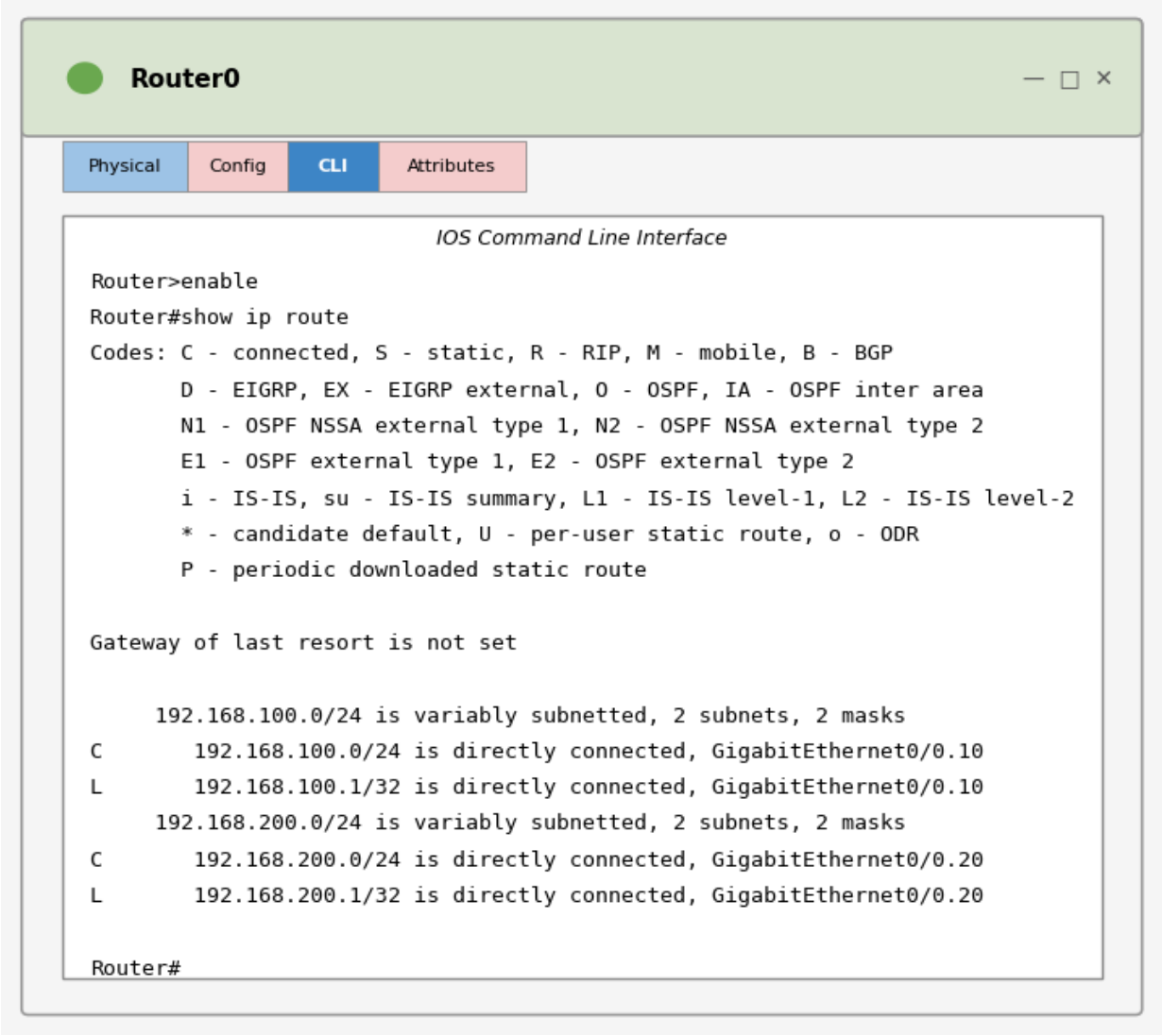
2. Router Interface Status (show ip interface brief)



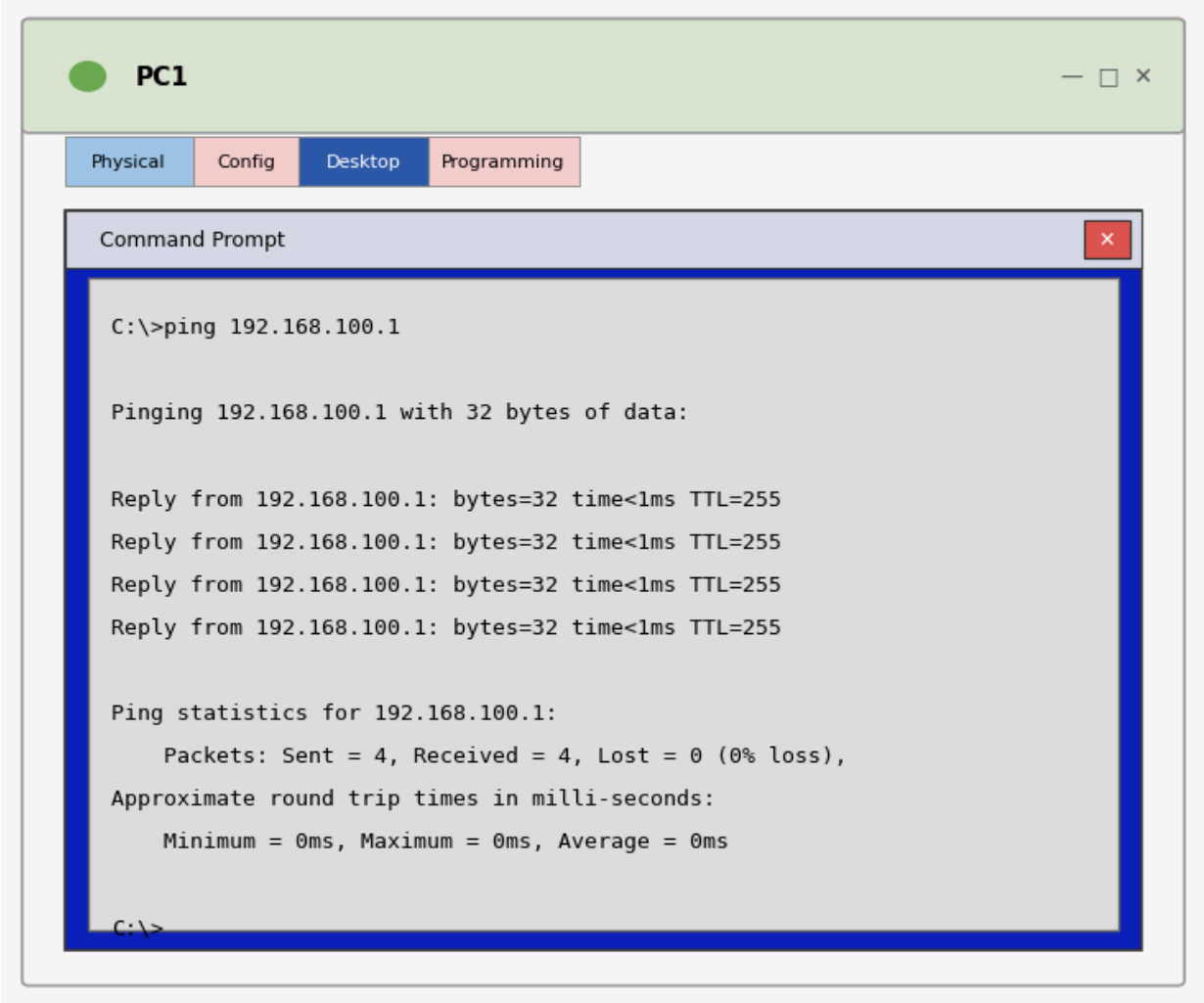
3. Trunk Status (show interfaces trunk)



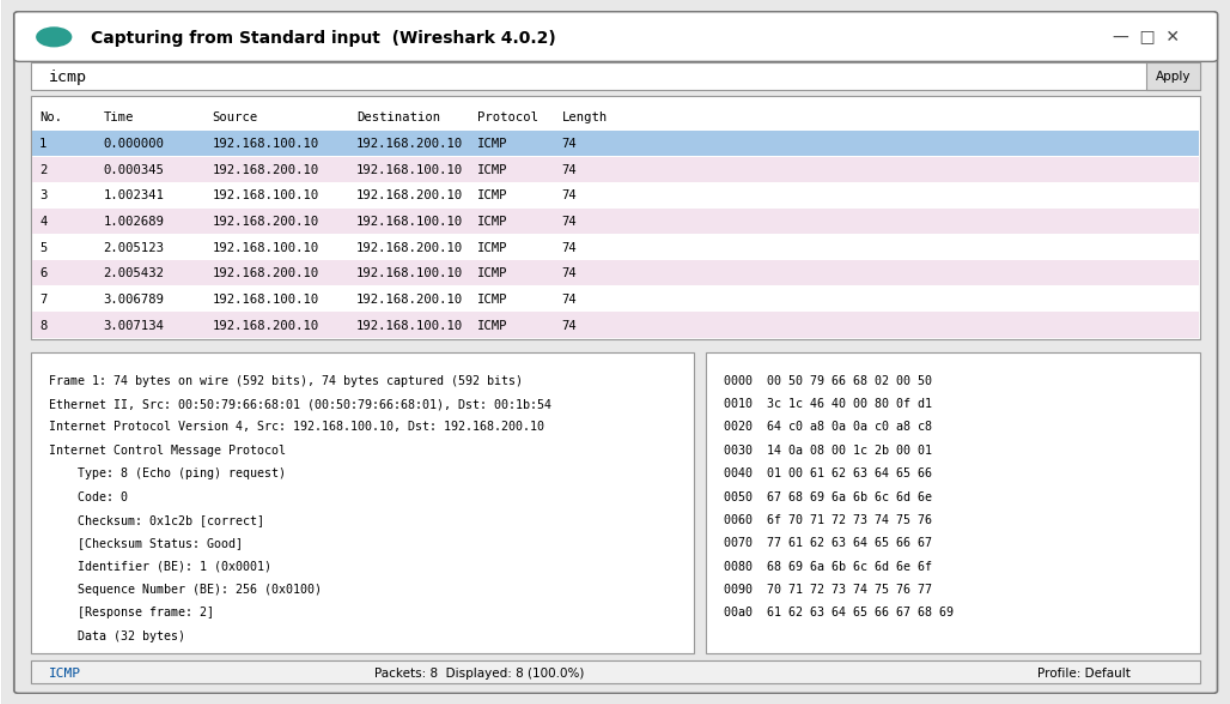
4. Routing Table (show ip route)



5. Ping to Default Gateway - PC1 to 192.168.100.1



6. ICMP Packet Capture (Wireshark)



7. Final Verification - Successful Inter-VLAN Ping to 192.168.200.10

